THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET B

Chapter 12: **Surface Areas and Volumes** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Use $\pi = \frac{22}{7}$ unless stated otherwise.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The curved surface area of a cone of radius 7 cm and slant height 10 cm is: 1

(a) 220 cm² **(b)** 154 cm² **(c)** 440 cm² **(d)** 110 cm²

2. Two solid hemispheres of the same radius r are joined along their bases. The volume of the solid formed is: 1

(a) $\frac{4}{3}\pi r^3$ **(b)** $\frac{2}{3}\pi r^3$ **(c)** $2\pi r^3$ **(d)** $4\pi r^3$

3. The total surface area of a solid cylinder of radius r and height h is: 1

(a) $2\pi r(r+h)$ **(b)** $2\pi rh$ **(c)** $\pi r(r+h)$ **(d)** $2\pi r^2 h$

4. The volume of a cone of radius 3 cm and height 7 cm is: 1

(a) 66 cm³ **(b)** 198 cm³ **(c)** 22 cm³ **(d)** 132 cm³

5. A student states: “The volume of a combined solid equals the sum of the volumes of its parts.” The correct response is: 1

(a) Correct
(b) Incorrect — volumes of parts can never be added
(c) Correct only for cubes
(d) Cannot be determined

6. A closed cylinder (radius 7 cm, height 10 cm) has a hemisphere of the same radius on top. Student A says its total surface area is 902 cm²; Student B says 1056 cm². Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

7. A cone and a hemisphere have equal bases and equal volumes. The ratio of the height of the cone to the radius is: 1

(a) 1 : 2 **(b)** 2 : 1 **(c)** 1 : 1 **(d)** 3 : 1

8. The volume of the largest sphere that can be carved from a cube of edge 7 cm is: 1

(a) 179.67 cm³ **(b)** 1437.33 cm³ **(c)** 343 cm³ **(d)** 89.83 cm³

9. The curved surface area of a hemisphere of radius 7 cm is: 1

(a) 308 cm² **(b)** 154 cm² **(c)** 462 cm² **(d)** 616 cm²

10. When a hemisphere is placed on a cylinder of the same radius, the surfaces not counted in the total surface area are: 1

(a) the curved surfaces
(b) the base of the hemisphere and the top of the cylinder
(c) the bottom of the cylinder
(d) none

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A solid is a cone (radius 6 cm, height 8 cm) mounted on a hemisphere of radius 6 cm. Find its total surface area. (Use $\pi = 3.14$) — 2

12. Find the volume of a solid formed by a cylinder of radius 2 cm and height 10 cm with a cone of height 3 cm (same radius) on top. — 2

13. Two cubes, each of volume 27 cm³, are joined end to end. A student says the volume of the cuboid is 27 cm³. Justify or refute, and find its surface area. — 2

14. A capsule is a cylinder with two hemispherical ends. Its total length is 21 mm and diameter 7 mm. Find its surface area. — 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A wooden toy is a cylinder of radius 3.5 cm and height 10 cm with a hemisphere of the same radius attached at each end. Find its total surface area and volume. — 3

16. A solid is a hemisphere of radius 2.1 cm surmounted by a cone of height 4 cm and the same radius. Find its volume. — 3

17. From a solid cylinder of height 2.4 cm and diameter 1.4 cm, a conical cavity of the same height and diameter is hollowed out. Find the total surface area of the remaining solid to the nearest cm². — 3

OR

A cone of radius 3 cm and height 4 cm is hollowed out from a solid cylinder of radius 3 cm and height 7 cm. Find the volume of the remaining solid. — 3

18. A bowl is a hollow hemisphere of internal radius 10.5 cm with a hollow cylinder of the same radius and height 14 cm on top. Find its inner surface area. — 3

OR

A hemisphere of radius 4.2 cm is scooped out from one end of a solid cylinder of radius 4.2 cm and height 10 cm. Find the total surface area of the remaining solid. — 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A building is in the form of a cylinder surmounted by a hemispherical dome and contains $41\frac{19}{21}$ m³ of air. If the internal diameter of the dome equals the total height of the building above the floor, find the height of the building. — 5

OR

A water tank is a cylinder with a hemispherical bottom. Its radius is 1.4 m and total depth 3.4 m. Find — 5
(i) its capacity in litres
(ii) the inner surface area to be painted.

20. **Case study:** A pencil is a cylinder of length 15 cm and radius 0.35 cm with a conical tip of height 1.2 cm. — 5
(i) Find the volume of the cylindrical part.
(ii) Find the volume of the tip.
(iii) Find the total volume.
(iv) Find the slant height and curved surface area of the tip.

OR

Case study: A circus tent is cylindrical to a height of 3 m and conical above it. The base radius is 52.5 m and the slant height of the cone is 53 m. — 5
(i) Find the area of canvas needed.
(ii) Find its cost at ₹12 per m².

END OF QUESTION PAPER — Total: 40 Marks